PROXIMAL BRACHIAL MONOMELIC AMYOTROPHY .

-RITHIKA M REDDY

-M S RAMAIAH MEDICAL COLLEGE.

- Stating the case is reported by ,I , Rithika consent of the patient taken prior to the examination .
- Author- Rithika M Reddy
- GUIDE- Dr. Akshay Rao.
- Department of Medicine.

CASE

- A 30 Year-old man who is farmer by occupation, residing in bagalkot presented with complaint of weakness in the proximal aspect of left upper limb since four (4) years.

HISTORY OF PRESENTING ILLNESS.

- Weakness in proximal aspect of left upper limb since 4 years; insidious onset, gradually progressive up until 6 months ago since when weakness is static.
- Difficulty in lifting heavy objects, lifting left arm above the shoulder.

- No pain, muscle cramps.
- No headache, neck pain, loss of consciousness.
- No numbness or sensory loss.
- No history of altered bowel or bladder functions.
- No history suggestive of cranial nerve deficits
- No trauma/ history of fever preceding the onset of weakness/ exposure to toxins or heavy metals.

PAST HISTORY

.No similar complaints in the past.

.No history of poliomyelitis, Diabetes Mellitus, Hypertension, epilepsy ,tuberculosis.

.No history of previous hospital admission or surgeries.

.No known allergies.

.No significant drug history.

FAMILY HISTORY

.No similar complaints in the family
.No significant family history.

PERSONAL HISTORY

Diet : vegetarian

Appetite : Normal

Sleep : undisturbed

Bowel and Bladder Movements : Regular

Non alcoholic , Non smoker

EXAMINATION.

- Young male , moderately built and nourished , conscious , cooperative and well oriented to time , place and person.
- VITALS: STABLE
- General physical examination is unremarkable.
- NUEROCUTANEOUS MARKERS:

SYSTEMIC EXAMINATION

- ***CENTRAL NERVOUS SYSTEM-***
- Higher Mental Functions normal.
- Fasciculations noted in the left deltoid.
- No winging of scapula.
- Decreased bulk in left arm.

RIGHT ARM CIRUMFERENCE	**LEFT ARM CIRCUMFERENCE**
29 cm	24 cm

- *TONE:*
- Decreased tone in the left arm.

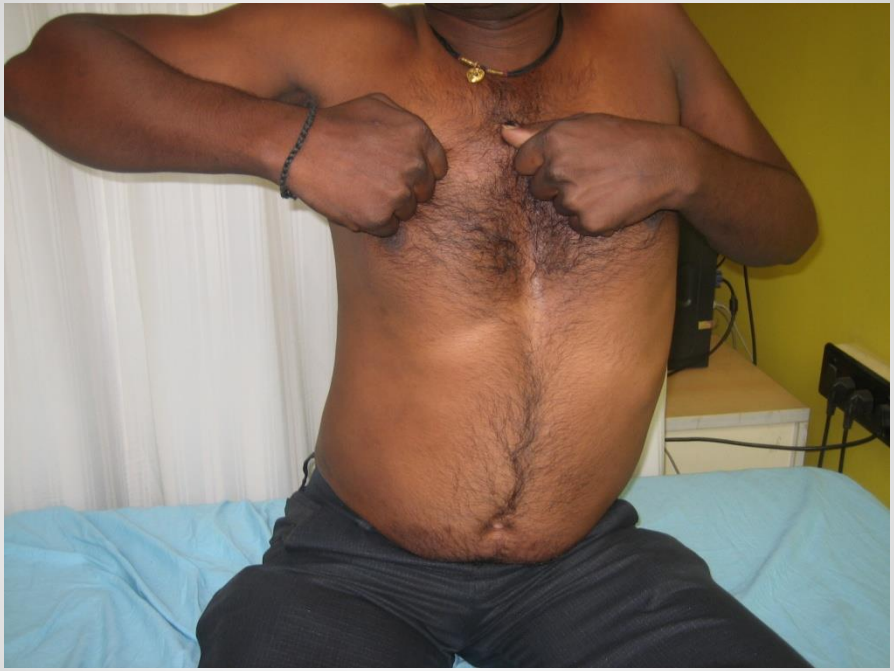
	LEFT	**RIGHT**
ARM:	Hypotonia.	Normal.
FOREARM:	Normal.	Normal.



.The above picture shows the loss of muscle bulk in the proximal aspect of the left upper limb.

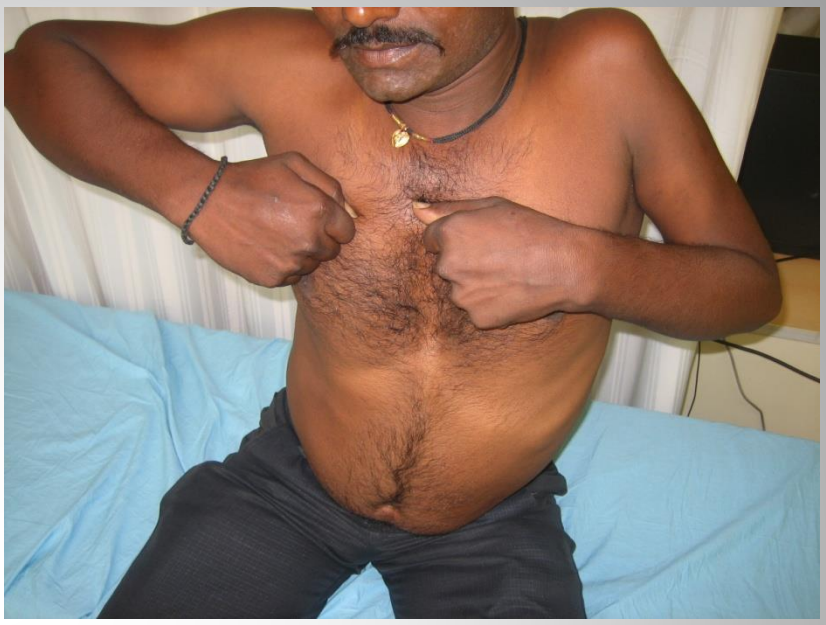
- *POWER*:

MUSCLES	RIGHT UPPER LIMB.	LEFT UPPER LIMB.
.DELTOID:	Normal	Weak
.BICEPS:	Normal	Weak
.TRICEPS:	Normal	Weak
.SUPINATOR:	Normal	Weak
.RHOMBOIDS:	Normal	Weak
.LATTISMUS DORSI:	Normal	Weak
.SMALL MUSCLES OF HAND :	Normal	Normal



WEAKNESS OF ABDUCTION.





*Photographs displaying atrophy of the proximal aspect of left upper limb with loss of contour of the left shoulder

MVI_2426.AVI

*THE ABOVE VIDEO SHOWS FASICULATIONS IN THE LEFT DELTOID.

- *REFLEXES*

REFLEXES	LEFT	RIGHT
.BICEPS JERK	1+	NORMAL
.TRICEPS JERK	1+	NORMAL
.SUPINATOR JERK	1+	NORMAL
.KNEE REFLEX	NORMAL	NORMAL
.ANKLE REFLEX	NORMAL	NORMAL

- Marked atrophy and weakness in proximal aspect of his left upper limb.
- Rest of the neurological examination was unremarkable.
- *Sensory , Cranial system examination*-Normal.

- *Cardiovascular system*- S1,S2 heard , no added murmurs.

- *Respiratory System*- Normal bilateral vesicular breath sounds , no added sounds.

- *Per abdomen*- No organomegaly noted, all quadrants move equally with respiration.

INVESTIGATIONS.

- Routine baseline investigations – within normal limits

- *MOTOR NERVE CONDUCTION STUDY*:
- Reduced compound motor action potentials (CMAP) amplitudes in nerves on the left side :
  - Axillary
  - Musculocutaneous
  - Infraspinatus

Procedure: Motor Nerve conduction was performed on bilateral Radial Axillary, Musculocutaneous, Supraspinatus and Infraspinatus nerves with elicitation of F-waves. Sensory nerve conduction was performed on bilateral Radial and Lateral cutaneous nerve of forearm and Medial cutaneous nerve of forearm nerves by using surface electrodes for stimulating and recording.

Observation:

Motor Nerve conduction study:

Normal distal motor latency, normal CMAP amplitude and normal motor conduction velocity on bilateral Radial nerves.
Normal distal motor latency, mildly reduced CMAP amplitude (compared to right) on left Axillary, Musculocutaneous and Infraspinatus nerves.

Normal distal motor latency, normal CMAP amplitude on right Axillary, Musculocutaneous, Supraspinatus and Infraspinatus nerves

Sensory nerve conduction study:

Normal sensory latency, normal SNAP amplitude and normal sensory conduction velocity on bilateral Radial, Lateral cutaneous nerve of forearm and Medial cutaneous nerve of forearm nerves

Impression:

Nerve conduction study shows left Axillary, Infraspinatus and Musculocutaneous nerve mildly reduced CMAP amplitude compared to right.

Name of C

Date of ad

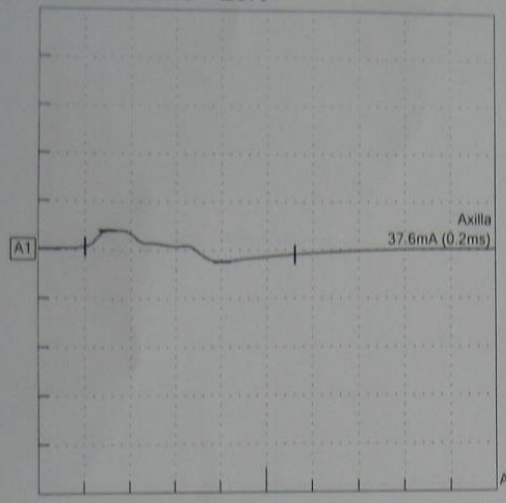
Discharg

Confirm

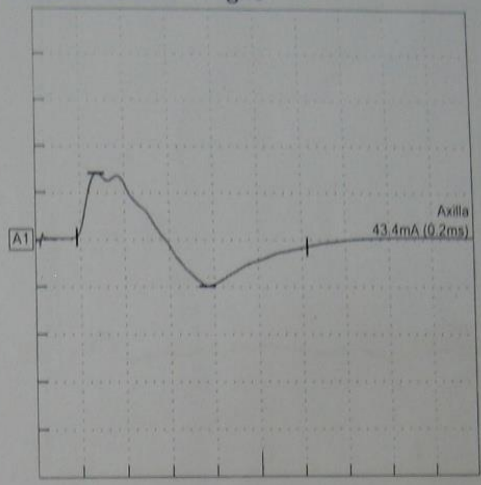
Pre-exi

Chief

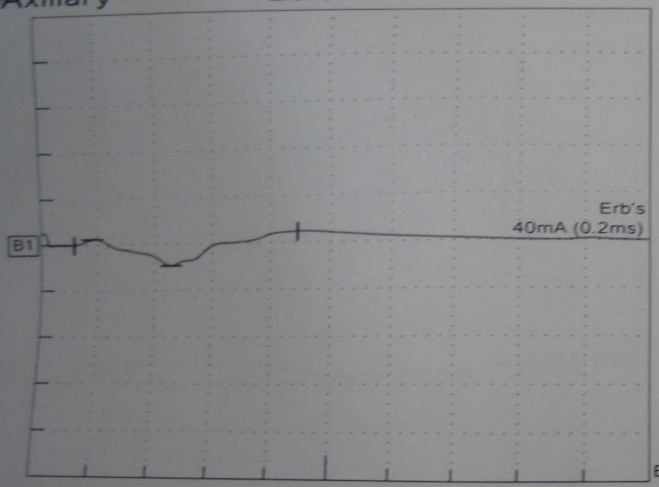
Musculocutane Left



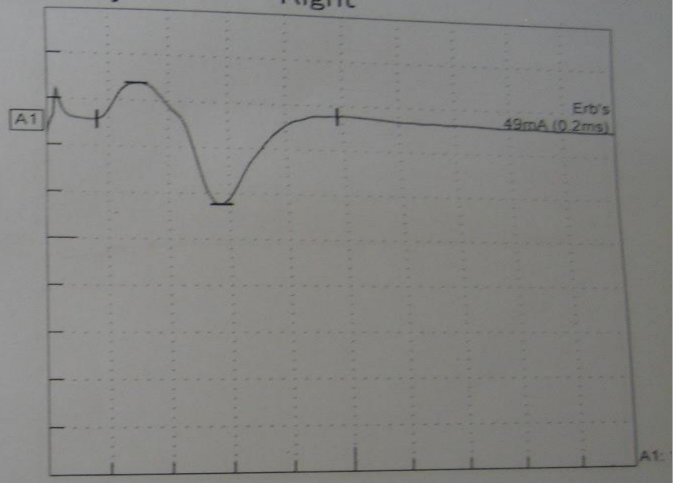
Musculocutane Right

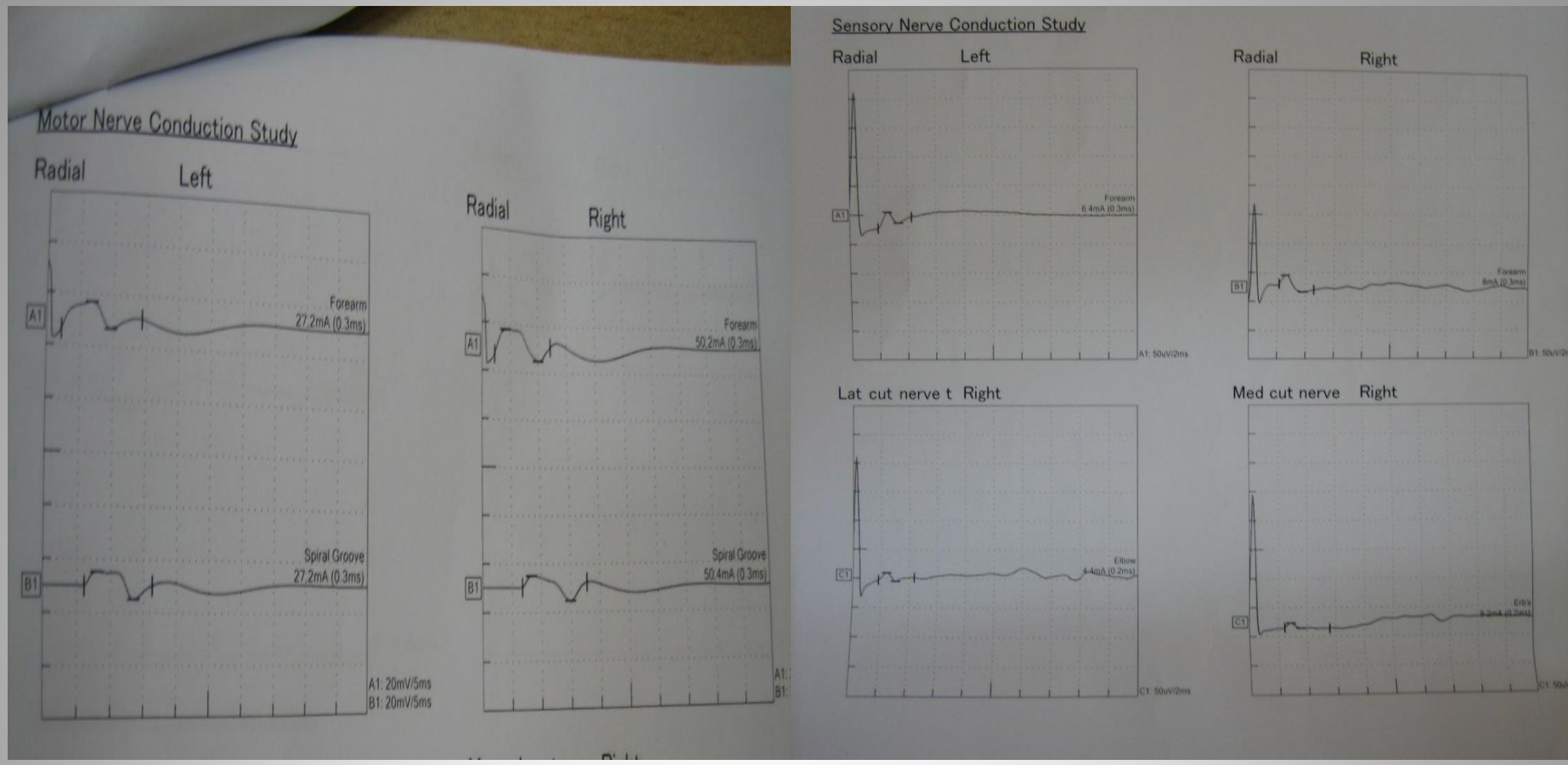


Axillary Left



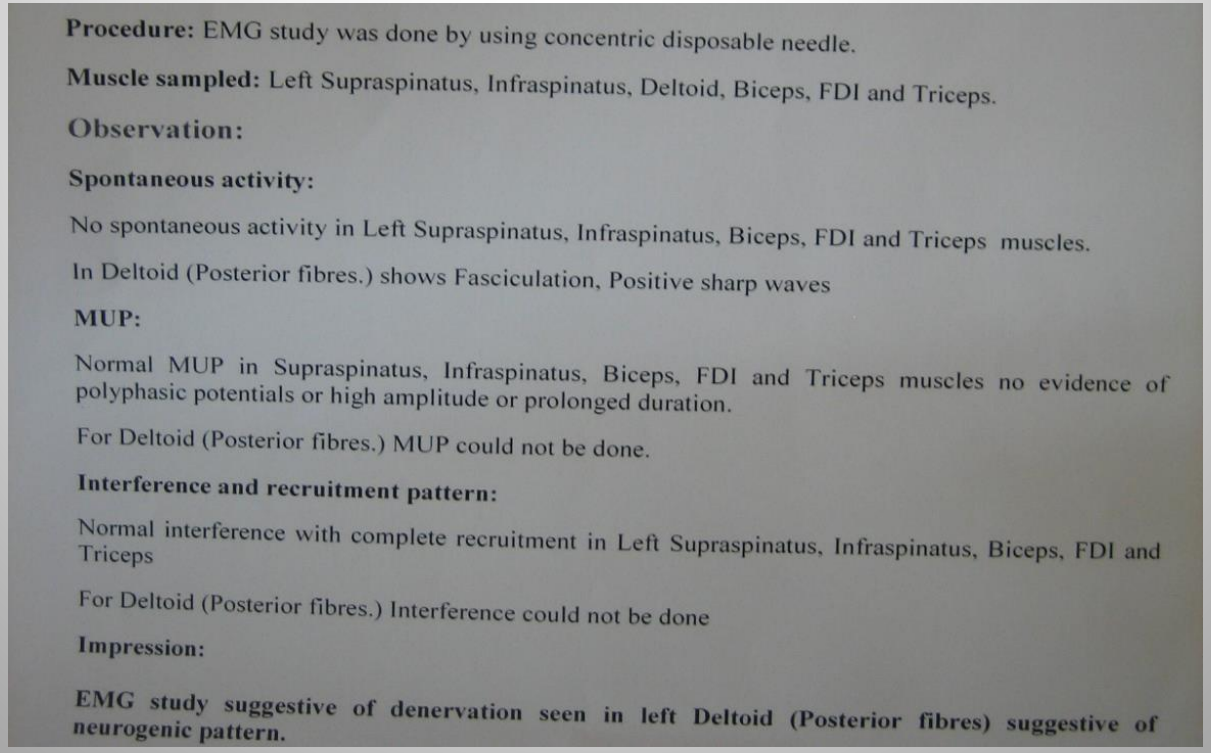
Axillary Right





Electroneurogram comparison of the nerves supplying upper limb muscles revealed reduced Compound Motor Action Potential (CMAP) on left musculocutaneous nerve.

- *ELECTRONEUROMYOGRAPHY (ENMG):*
- Denervation pattern with ↓ motor action potentials.
- Fasiculations in left deltoid.



Procedure: EMG study was done by using concentric disposable needle.

Muscle sampled: Left Supraspinatus, Infraspinatus, Deltoid, Biceps, FDI and Triceps.

Observation:

Spontaneous activity:

No spontaneous activity in Left Supraspinatus, Infraspinatus, Biceps, FDI and Triceps muscles.

In Deltoid (Posterior fibres.) shows Fasciculation, Positive sharp waves

MUP:

Normal MUP in Supraspinatus, Infraspinatus, Biceps, FDI and Triceps muscles no evidence of polyphasic potentials or high amplitude or prolonged duration.

For Deltoid (Posterior fibres.) MUP could not be done.

Interference and recruitment pattern:

Normal interference with complete recruitment in Left Supraspinatus, Infraspinatus, Biceps, FDI and Triceps

For Deltoid (Posterior fibres.) Interference could not be done

Impression:

EMG study suggestive of denervation seen in left Deltoid (Posterior fibres) suggestive of neurogenic pattern.

- *MRI CERVICAL SPINE:*
- Cervical spine contrast MRI revealed mild loss of cervical lordosis.





MRI of cervical spine in flexion – absence of typical features of Hirayama disease

TREATMENT.

- The patient was managed with supportive care and has been under regular follow up ever since.

- Clinical status has been stable.

CASE REPORT

-BRACHIAL MONOMELIC AMYOATROPHY

-ISOLATED ATROPHY , WEAKNESS AND FASCICULATION IN PROXIMAL LEFT UPPER LIMB.

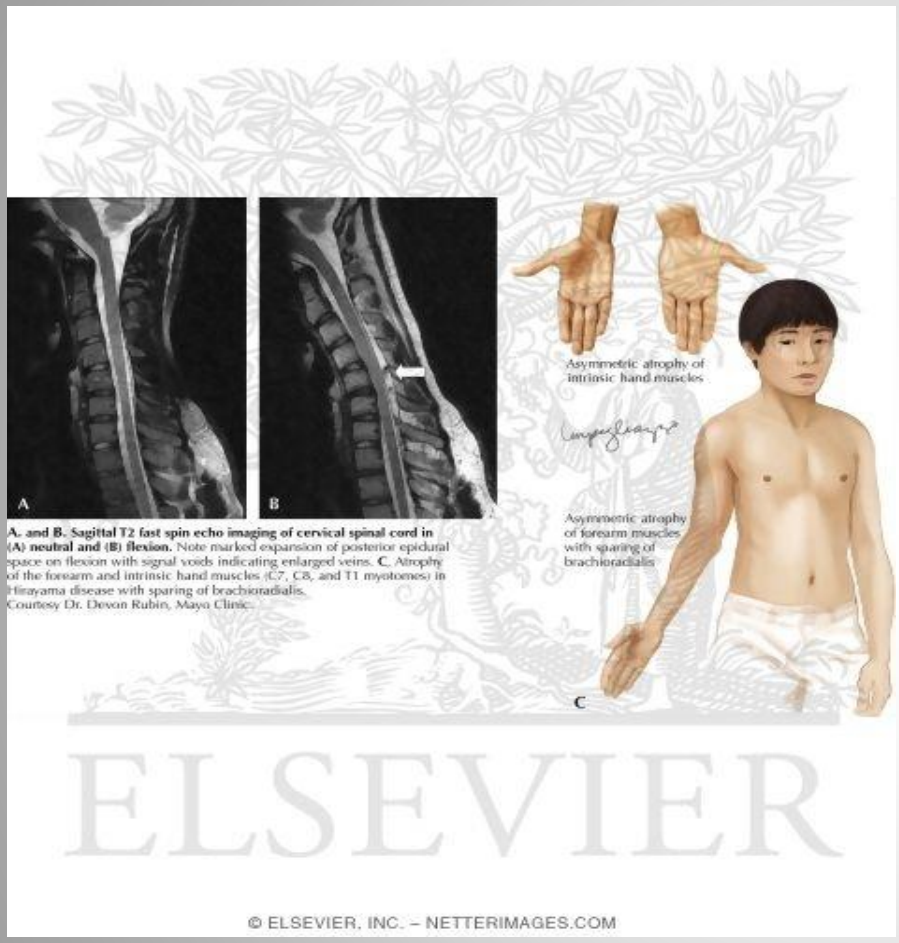
-WITHOUT TYPICAL CERVICAL SPINE FLEXION MRI FINDINGS OF HD.

BACKGROUND-

- BMMA/HD initially recognized –Japan,1959.
- Juvenile asymmetric segmental spinal muscular atrophy.
- Benign focal amyotrophy involving distal aspect of upper limb - flexion induced cervical myelopathy as evidenced on MRI of cervical spine on flexion studies

DISCUSSION.

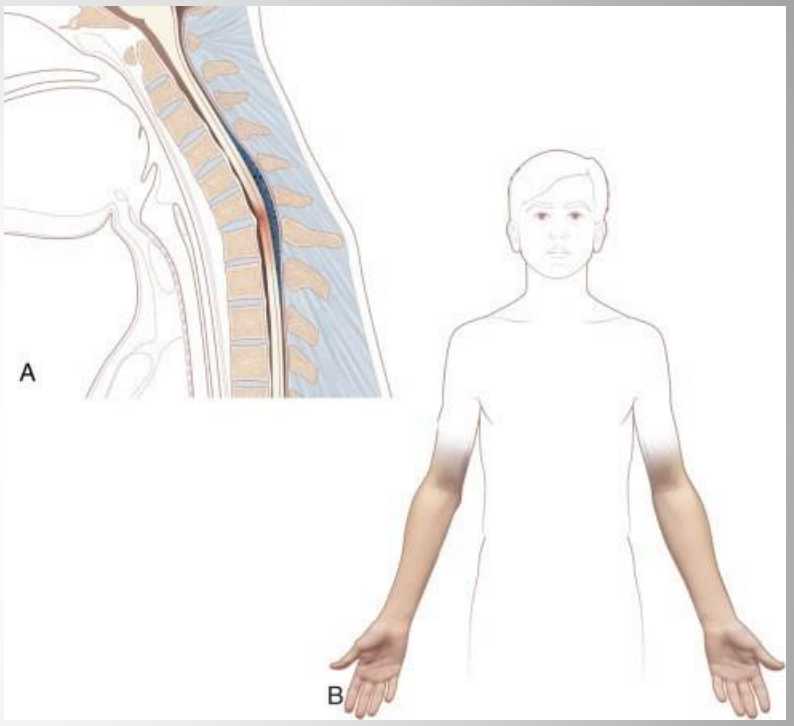
- Hirayama disease = BMMA ?
- Sporadic disease.
- Insidious onset, progression for 2-4 years ,followed by stationary course.
- Unilateral distal upper limb weakness and wasting.
- Focal degenerative motor neuron disease of cervical cord- Variant of MND?

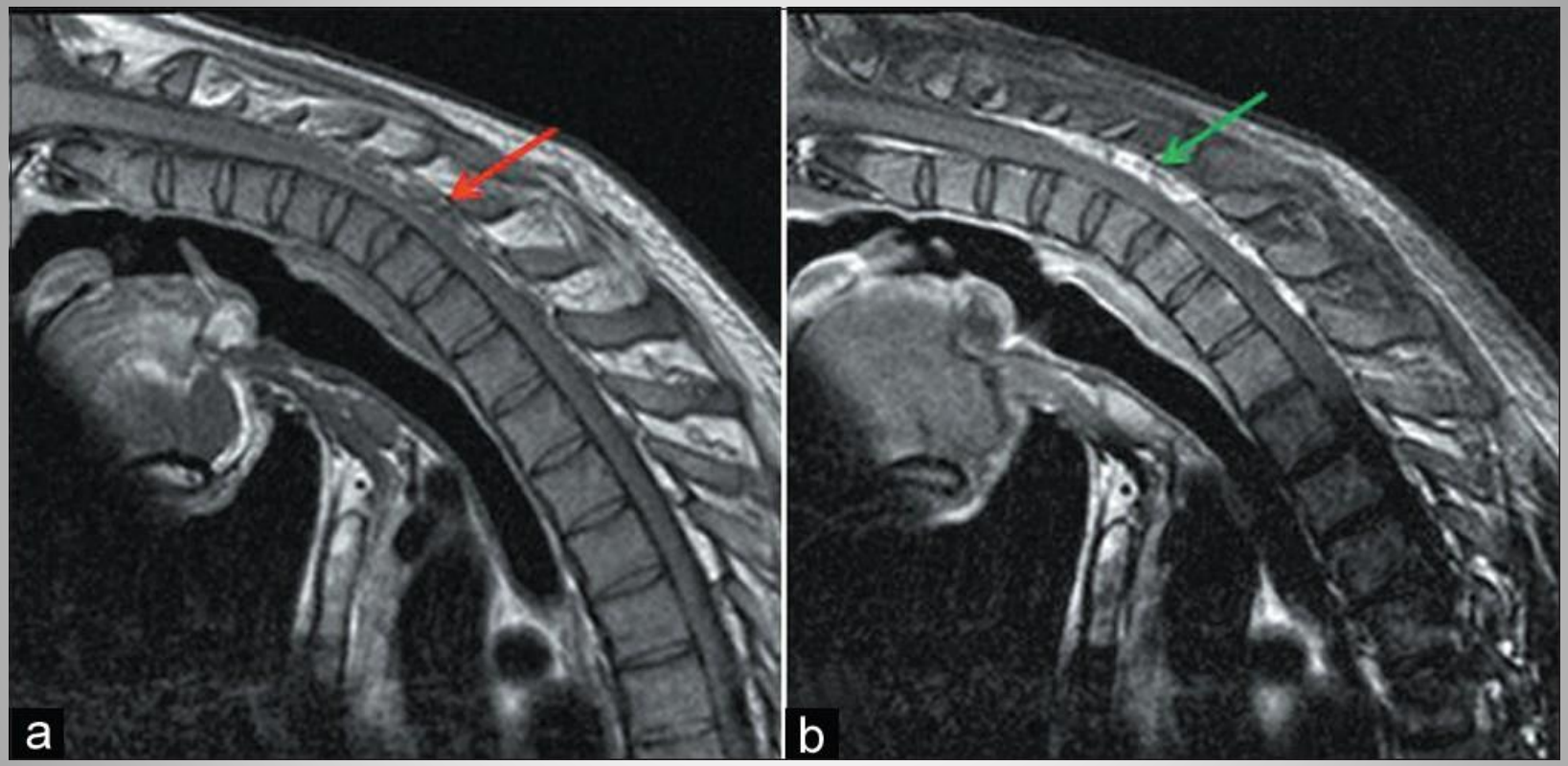


A. and B. Sagittal T2 fast spin echo imaging of cervical spinal cord in (A) neutral and (B) flexion. Note marked expansion of posterior epidural space on flexion with signal voids indicating enlarged veins. **C.** Atrophy of the forearm and intrinsic hand muscles (C7, C8, and T1 myotomes) in Hirayama disease with sparing of brachioradialis. Courtesy Dr. Devon Rubin, Mayo Clinic.

ELSEVIER

© ELSEVIER, INC. – NETTERIMAGES.COM





. flexion contrast MRI showing epidural crescentic enhancing region (green arrow)

- No cranial nerve, cerebral, pyramidal tract or sensory involvement.
- EMG- denervation pattern of fibrillations, positive sharp waves and fasiculations
- MRI abnormalities of cervical spinal cord on flexion studies hence ?

  Flexion induced cervical myelopathy

.Imbalanced growth.

⇩

.Disproportional length between patients vertebral column and spinal canal.

⇩

.Tightening of dural sac
.Anterior displacement of posterior dural wall.

⇩

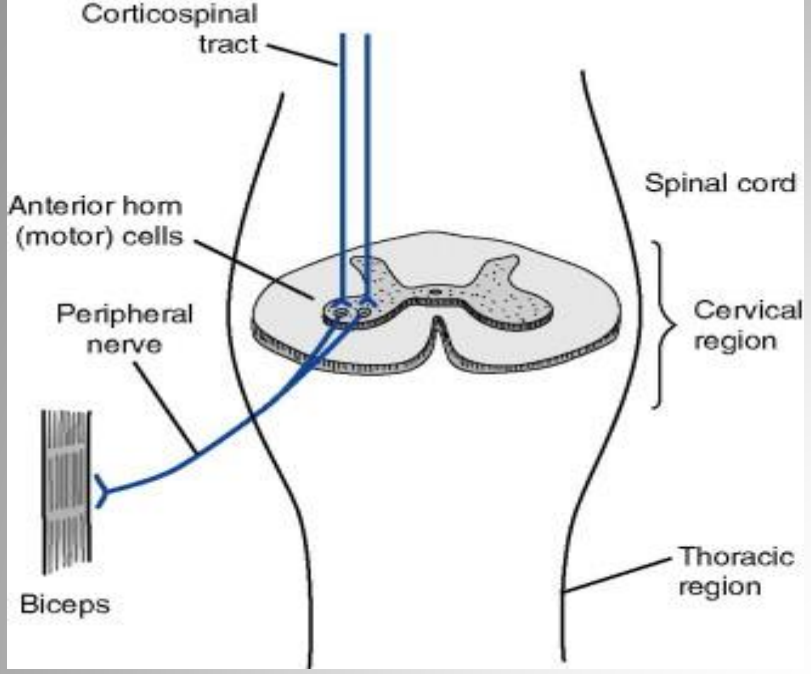
.Compressing cord against the post. margin of adjacent vertebrae.

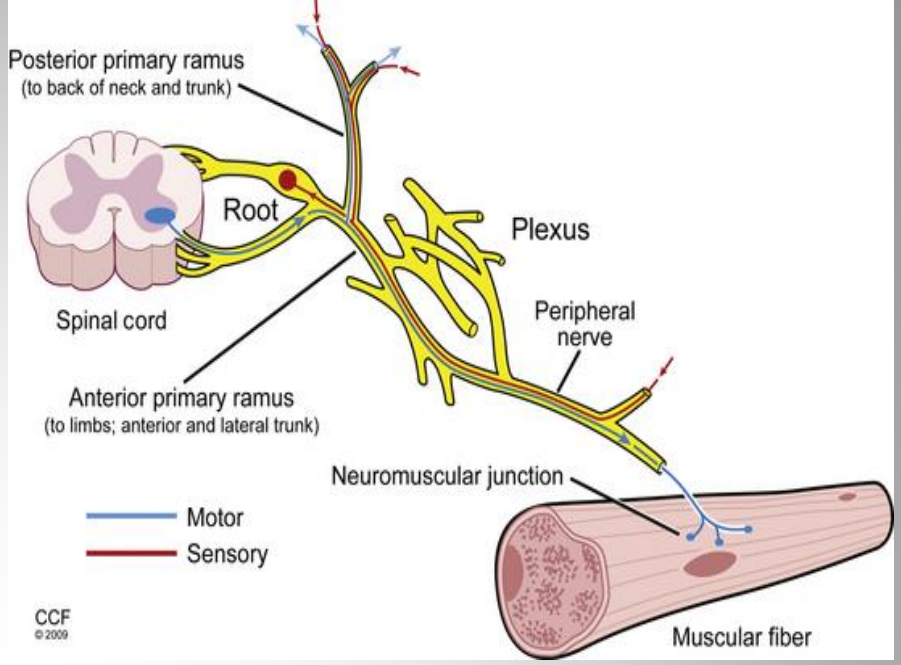
⇩

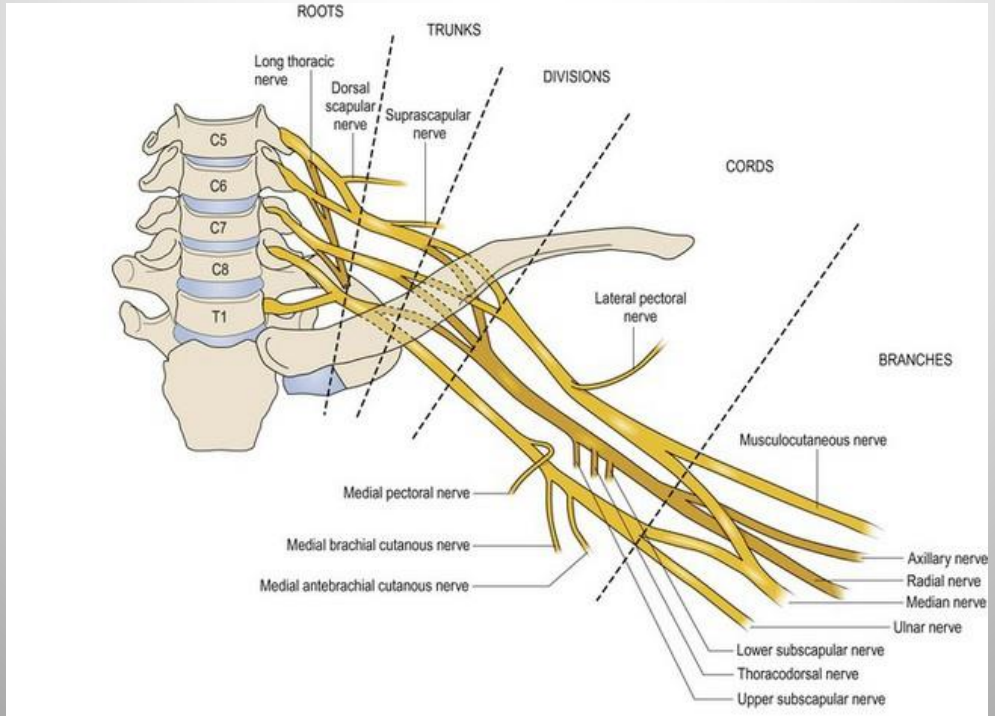
.Microcirculatory disturbances in cervical cord

⇩

.Necrosis of AHC(lower cervical cord atrophy.


Corticospinal tract
Spinal cord
Anterior horn (motor) cells
Peripheral nerve
Cervical region
Thoracic region
Biceps


Posterior primary ramus
(to back of neck and trunk)
Root
Plexus
Spinal cord
Peripheral nerve
Anterior primary ramus
(to limbs; anterior and lateral trunk)
Neuromuscular junction
Motor
Sensory
CCF
© 2009
Muscular fiber


ROOTS
TRUNKS
DIVISIONS
CORDS
BRANCHES
Long thoracic nerve
Dorsal scapular nerve
Suprascapular nerve
C5
C6
C7
C8
T1
Lateral pectoral nerve
Musculocutaneous nerve
Medial pectoral nerve
Medial brachial cutanous nerve
Medial antebrachial cutanous nerve
Axillary nerve
Radial nerve
Median nerve
Ulnar nerve
Lower subscapular nerve
Thoracodorsal nerve
Upper subscapular nerve

- HD = BMMA?
- “HIRAYAMA DISEASE” be used for unilateral upper limb **distal** muscle weakness and atrophy + cervical spine MRI findings.
- Restricting BMMA – **Proximal** upper limb atrophy without typical MRI features of HD.

DIAGNOSIS.

- PROXIMAL BRACHIAL MONOMELIC AMYOTROPHY (PMMA) .

DIFFERENTIAL DIAGNOSIS:

Unilateral upper limb weakness & atrophy

- Multifocal motor neuropathy with conduction blocks.
- Spinal muscular atrophy.
- Motor neuropathy.
- Chronic focal myositis.
- Brachial neuritis.
- Late –progression of poliomyelitis.

.SPINAL MUSCULAR ATROPHY AND MOTOR NEUROPATHY-

.Upper limbs involvement is **bilateral** and **symmetric**.

.NEURALGIC AMYOTROPHY OR BRACHIAL NEURITIS-

Deep shoulder **aching dysaesthesia**

.Good prognosis.

.MULTIFOCAL MOTOR NEUROPATHY WITH CONDUCTION BLOCKS-

.Upper limbs asymmetric predominant **distal** muscle involvement

ENMG shows **demyelination with conduction blocks**

High titers of anti- GMI ganglioside antibodies.

.CHRONIC FOCAL MYOSITIS-

EMG – **myopathic** pattern

↑ serumcreatine phosphokinase.

.LATE –PROGRESSION OF POLIOMYELITIS

.Past poliomeylitis

PREVALENCE

- Proximal brachial MMA very limited literature, and none have shown dynamic compression of cervical cord on MRI.
- In 2000 –MRG published case of PMMA with normal cervical MRI.
- In 2008- case of two young men with PMMA , normal cervical MRI.
- In 2013 – Hassan reported case of two young men with PMMA , mild loss of cervical lordosis on MRI.

CONCLUSION

- Consider **proximal** BMMA as a separate entity whereas Hirayama disease - **distal** upper limb involvement with cervical MRI findings on flexion studies.
- Need to explore pathogenesis for BMMA.
- Not indicated to perform flexion contrast MRI of the cervical spine in **proximal** BMMA patients.
- This case unlike HD , unilateral proximal upper limb muscle weakness with denervation atrophy isn’t a form of flexion induced cervical myelopathy.

REFERENCES

- Hirayama K et al, Juvenile muscular atrophyof unilateral upper extremity: A new clinical entity. PsychiatrNeurol Jpn.1959;61:2190-7
- Uncini A, Servidei S, DelliPizzi C, Cutarella R, Di Muzio A, Gambi D, etal.Benignmonomelicamyotrophy of lower limb: report of three cases. ActaNeurol Scand. 1992;85:397-400https://www.ncbi.nlm.nih.gov/pubmed/1642110
- Hassan KM, Sahni H. Nosology of juvenile muscular atrophy of distal upper extremity: from monomelic amyotrophy to Hirayama disease--Indian perspective. Biomed Res Int. 2013;2013:478516https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3770029/
- De Frietas Marcos R. G., Nascimento Osvaldo J.M., BenignMonomelic Amyotrophy: A study of 21 cases; Arq Neuropsiquiatr.2000; 58(3-B):808-813https://pdfs.semanticscholar.org/4da3/405f8bf18155c0b988f1dea56d2fe26d8b8e.pdf
- Aundhakar SC, Mahajan SK, Chhapra DA. Hirayama's Disease: A Rare Clinical Variant of Amyotrophic Lateral Sclerosis. Adv Biomed Res. 2017;6:95. Published 2017 Jul 28. doi:10.4103/2277-9175.21179https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5549545/
- Yoo SD, Kim HS, Yun D H, Kim DH, Chon J, LeeSA, et al. Monomelicamyotrophy (hirayama disease) with upper motor neuron signs: a case report. Ann Rehabil Med. 2015;39(1):122-7.https://www.ncbi.nlm.nih.gov/pubmed/25750881